

Original contribution

Hemodynamic investigation of peritumoral impaired blood oxygenation-level dependent cerebrovascular reactivity in patients with diffuse glioma

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ABSTRACT

Introduction: The presence of peritumorally impaired blood oxygenation-level dependent cerebrovascular reactivity (BOLD-CVR) has been unequivocally demonstrated in patients with diffuse glioma, and may have value to better identify tumor infiltration zone. Since BOLD-CVR does not measure hemodynamic changes directly, we performed additional MR perfusion studies to better characterize the peritumoral hemodynamic environment. **Methods:** Seventeen patients with WHO grade III and IV diffuse glioma underwent high resolution advanced hemodynamic MR imaging including BOLD-CVR and MR perfusion. The obtained multiparametric hemodynamic factors (i.e., regional cerebral blood flow (rCBF), relative cerebral blood volume (rCBV), mean transit time (MTT), time-to-peak (TTP) and BOLD-CVR, were analyzed within 10 concentric expanding 3 mm volumes of interest (VOIs) up to 30 mm from the tumor tissue mask.

Results: BOLD-CVR impairment was found within the tumor tissue mask and the peritumoral VOIs up to 21 mm as compared to the contralateral flipped CVR analysis ($p < 0.05$). In the affected hemisphere, we observed positive spatial correlations including all VOIs between BOLD-CVR and rCBV values ($r = 0.27$; $p < 0.001$), rCBF ($r = 0.42$; $p < 0.001$) and a negative correlation between BOLD-CVR and TTP ($r = -0.47$; $p < 0.001$).

Conclusions: Peritumorally impaired BOLD-CVR is associated with concomitant hemodynamic alterations with severity correlating to tumor volume. The distribution of these multiparametric hemodynamic MRI patterns may be considered for future studies characterizing the hemodynamic peritumoral environment, thereby better identifying the extent of tumor infiltration.

1. Introduction

The presence of peritumorally impaired blood oxygenation-level dependent cerebrovascular reactivity (BOLD-CVR) has been unequivocally demonstrated in patients with diffuse glioma [1–5]. In particular, previous studies have highlighted the importance of impaired BOLD-CVR mapping in relationship to neurovascular uncoupling [3]. Such an analysis may aid in identifying eloquent brain areas, for instance, in presurgical planning of tumor resection [3]. The exact underlying pathophysiology of impaired BOLD-CVR, however, is yet to be investigated as the BOLD fMRI signal contains both hemodynamic and metabolic components [6–8].

MR perfusion-weighted imaging (MRP) is a contemporary clinical imaging tool for newly diagnosed and recurrent diffuse gliomas, as it allows for a multiparametric hemodynamic assessment, including regional cerebral blood flow (rCBF), relative cerebral blood volume (rCBV), mean transit time (MTT), and time-to-peak (TTP) [9].

In particular, for the contrast enhancing lesion on T1-weighted MRI it supports identifying malignant glioma tissue [10–13]. However, the specific hemodynamic alterations beyond the contrast-enhancing lesion (i.e., peritumoral tissue) have not been well studied.

We, therefore, aim to characterize the peritumoral hemodynamic environment better using BOLD-CVR and high-resolution MR perfusion.

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Table 1

Patient & tumor characteristics (MRP: magnetic resonance perfusion; BOLD: blood oxygenation-level dependent; GBM: glioblastoma; AA: anaplastic astrocytoma; AO: anaplastic oligodendroglioma).

Patient	Age/sex	Days between MRP & BOLD examination	Diagnosis/WHO Grade	IDH1 ^{R132H} status	Tumor volume (cm ³)
1	58/F	0	GBM/IV	Wildtype	48.06
2	72/M	0	GBM/IV	Wildtype	40.98
3	72/M	11	GBM/IV	Wildtype	17.56
4	74/M	3	GBM/IV	Wildtype	27.58
5	54/M	2	GBM/IV	Wildtype	5.73
6	32/F	13	AA/III	Mutant	1.21
7	56/M	5	GBM/IV	Mutant	86.18
8	56/M	12	AO/III	Wildtype	12.63
9	71/M	12	GBM/IV	Wildtype	7.45
10	48/M	31	GBM/IV	Mutant	0.51
11	58/F	1	AO/III	Mutant	0.55
12	65/M	3	GBM/IV	Wildtype	4.33
13	53/M	0	AA/III	Wildtype	200.34
14	27/M	8	GBM/IV	Wildtype	44.43
15	52/M	1	GBM/IV	Mutant	9.67
16	60/F	3	AA/III	Mutant	7.46
17	61/M	5	GBM/IV	Wildtype	56.11

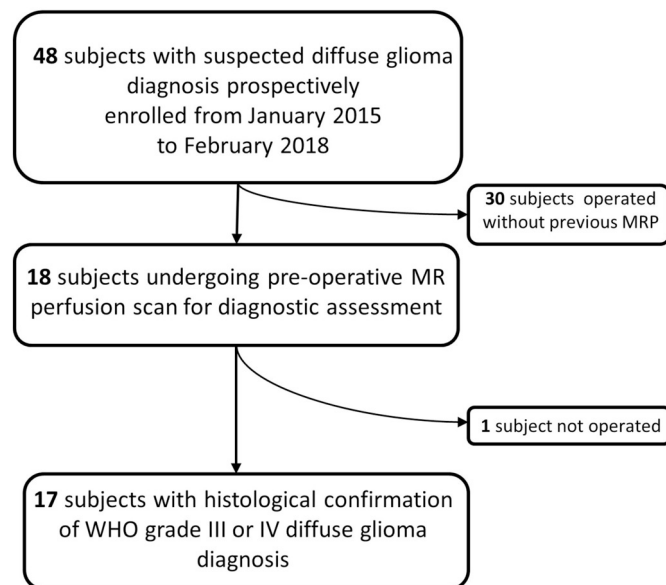


Fig. 1. Flow-chart describing patients included in the study.

Perilesional hemodynamic parameters were measured within ten concentric expanding 3 mm volumes of interest (VOIs) starting from the tumor tissue mask. This peritumoral hemodynamic analysis fostered specific patterns that may be entertained to better study the extent of tumor infiltration.

2. Methods

2.1. Patients selection

We selected patients with WHO grade III and IV diffuse glioma from our ongoing prospective BOLD-CVR glioma database, starting in January 2015–February 2018 (research protocol KEK-ZH-No. 2012-0427), who underwent both a preoperative BOLD-CVR and a 3-Tesla MRP examination within a maximum time interval of 1 month (Table 1). Histopathological diagnosis was available in all patients (Fig. 1). We identified twelve patients diagnosed with glioblastoma, three patients with anaplastic oligodendroglioma (WHO grade III) and two patients with anaplastic astrocytoma WHO grade III (Table 1).

2.2. MRI acquisition protocol

Studies were acquired on a 3-Tesla Skyra VD13 (Siemens, Erlangen, Germany) using a 32-channel head coil on a sagittal image with voxel size: $3 \times 3 \times 3$ mm³, slice gap 0.3 mm. A high-resolution 3D T1-weighted anatomical image was acquired with the same orientation as the BOLD fMRI scan (voxel size: $0.8 \times 0.8 \times 1.0$ mm³). BOLD fMRI parameters: axial two dimensional (2D) single-shot EPI sequence planned on the ACPC line plus 20° (on a sagittal image) voxel size $3 \times 3 \times 3$ mm³, acquisition matrix $64 \times 64 \times 35$ ascending interleaved slice acquisition, slice gap 0.3 mm, GRAPPA factor 2 with 32 reference lines, repetition time (TR)/echo time (TE) 2000/30 ms, flip angle 85°, bandwidth 2368 Hz/Px, Field of View 192×192 mm². Further acquisition method parameters can be found in a previous publication [14].

The sequence parameters of the dynamic susceptibility contrast (DSC) T2*-perfusion were: total time of acquisition (TA) 1:46 min:s, 60 volumes, field of view (FoV) 220×220 mm², 25 transversal slices of 4.0 mm thickness with a 30% distance factor, phase encoding direction A > P, voxel size $1.7 \times 1.7 \times 4.0$ mm³, single shot acquisition resolution 128×128 , GRAPPA factor 3, 60 reference lines, TR/TE 1600/27.0 ms, flip angle 90°, fat suppression, standard B0 shim, TrueForm B1 Shim, echo spacing 0.94 ms, receive bandwidth 1220 Hz/Px.

Estimation of the perfusion maps: rCBF: [ml/100 g]/min, rCBV: ml/100 g, MTT: seconds, and TTP: seconds, was performed off-line using OLEA Sphere (Version 3.0-SP6, Olea Medical SA, La Ciotat, France). These multiparametric maps, defined under the standard perfusion model hypothesis, are derived using automatic vascular input determination and deconvolution.

2.3. Standardized carbon dioxide (CO₂) stimulus application

BOLD-CVR was studied by applying an iso-oxic CO₂ stimulus with a custom-build computer-controlled gas blender (RepirAct™, Thornhill Research Institute, Toronto, Canada) containing the prospective gas targeting algorithms. BOLD signal changes were induced by a single hypercapnia pseudo-square wave CO₂ change. The pseudo-square wave consisted of a hypercapnia plateau of ~10 mmHg CO₂ from the previously registered subject's CO₂ baseline for 80 s, followed by a return to the baseline CO₂ level. Further details of this technique have been reported in a previous publication [14].

2.4. Data analysis

All images were preprocessed using Statistical Parameter Mapping software (SPM 12, Wellcome Trust Centre for Neuroimaging, Institute of Neurology, University College London; <http://www.fil.ion.ucl.ac.uk/spm/>). Slice timing correction was applied to correct for the interleaved acquisition. The T1-weighted image was linearly registered to the mean BOLD volume for optimal alignment. Automated segmentation of the T1-weighted image yielded gray and white matter, cerebrospinal fluid, skull, and skin probability maps.

Further processing was done using MATLAB2013 based (The MathWorks, Inc., Natick, United States; www.mathworks.com) in-house written scripts and a previously published analysis method [14]. In brief, BOLD-CVR was calculated by a linear least square fit between the BOLD time series to the carbon dioxide time series on a voxel-per-voxel basis and was defined as the %BOLD signal change per mmHg CO₂. For overlay purposes, BOLD-CVR was color-coded within a range of −0.6% and 0.6% BOLD signal change/mmHg CO₂ and overlaid on the T1-weighted image. Similar overlay maps were created for rCBV, rCBF, MTT and TTP (Fig. 2).

2.5. Volume of interest determination

Tumor masks were manually drawn using iPlan software (BrainLab

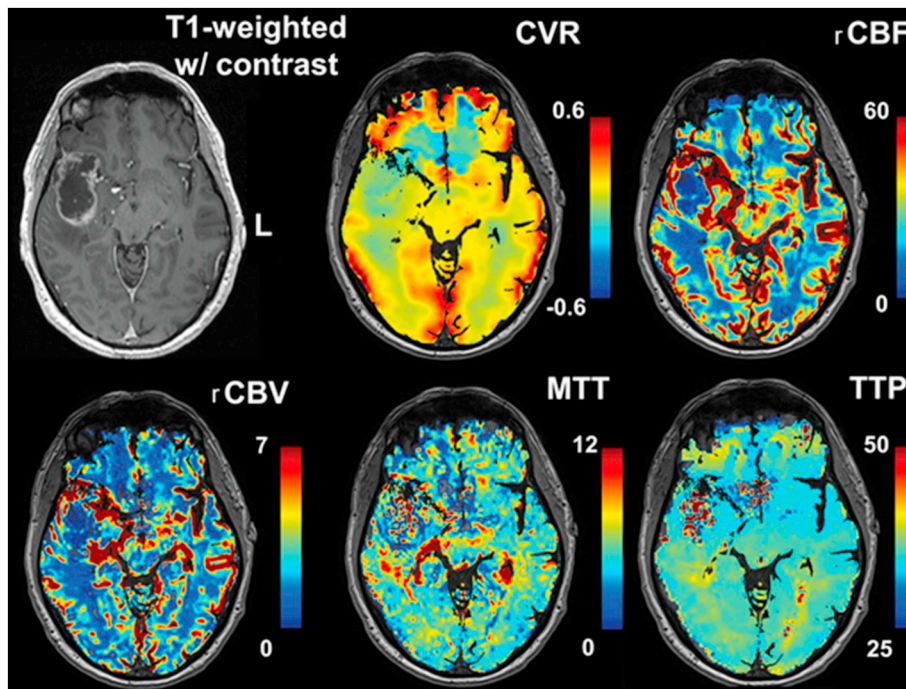


Fig. 2. Example of high-resolution BOLD-CVR and MR Perfusion imaging in a patient with a right temporal glioblastoma (IDH1^{R132H} wildtype, ATRX nuclear expression). BOLD-CVR and MR perfusion maps used as an overlay on an anatomical axial T1-weighted volume showing a right temporal lesion with later histological confirmation of a glioblastoma. Qualitatively, one can appreciate marked CVR, rCBF and rCBV impairment within the contrast-enhancing tumor lesion. Concerning the tumor border and perilesional tissue evident hemodynamic alterations can be seen. However, a more detailed quantitative analysis is warranted to interpret better the pathophysiology (as presented in this paper). CVR: cerebrovascular reactivity (color scale: % Δ BOLD signal/ Δ mmHgCO₂); rCBF: relative cerebral blood flow (color scale: [ml/100g]/min); CBV: cerebral blood volume (color scale: ml/100g); MTT: mean transit time (color scale: seconds); TTP: time to peak (color scale: seconds); L: left).

AG, Munich, Germany) on T1-weighted contrast-enhanced scans for every slice on which the tumor was visible after merging with the anatomical non-contrast enhanced T1-weighted scan. This allowed obtaining a 3-dimensional (i.e., spherical) volumes of interest (VOIs) of the contrast-enhancing tumor lesion. These tumor masks were overlaid on the BOLD-CVR and MRP maps to obtain mean intra-lesional values in those specific volumes.

Analysis of the peritumoral tissue was done by creating automated 3-dimensional (x-y-z-axis) concentric VOIs of 3 mm width expanding from the tumor border outwards up to 30 mm (see Fig. 3). These concentric expanding volumes were overlaid on the BOLD-CVR and MRP maps values, as described previously [15]. Subsequently, tumor volume, whole brain, gray matter, and white matter mean BOLD-CVR and MRP values were obtained. To confirm impaired peritumoral BOLD-CVR, the same analysis was performed on the contralateral hemisphere in mirror-like anatomical regions (i.e., a flipped analysis; Fig. 3). The two hemispheres were analyzed separately in order to avoid overlap.

2.6. Statistical analysis

All statistical analyses were performed with MATLAB2016 and SPSS Statistics 23 (IBM Corp. Armonk, NY, USA). Shapiro-Wilk test was performed on raw data to assess for normal distribution. Each VOI was analyzed separately. P-values < 0.05 were considered significant. As each variable showed a normal distribution, the comparison between affected and healthy hemisphere in each sphere was made for BOLD-CVR with a paired t-test. The Pearson's correlation coefficient was determined to find a correlation between mean BOLD-CVR and MRP parameter values in the different volumes of interest: this analysis was performed on the whole cohort. A partial correlation was run to determine the relationship between BOLD-CVR and MRP parameters while correcting for the tumor volume. A weighted least-squares regression was performed to assess the influence of the independent variable tumor volume, WHO grading, and IDH status on CVR and MRP inside the lesion and in the studied VOIs. To correct for multiple comparisons, a Benjamini-Hochberg step-up procedure [16] and a Benjamini, Krieger & Yekutieli procedure [17] were performed for independent and dependent comparisons, respectively, with a false discovery rate of 0.05.

3. Results

3.1. Patients and tumor features

A flowchart describing the patient selection is depicted in Fig. 1. Seventeen patients with a WHO grade III and IV diffuse glioma were included, 13 were males (72%). Mean age was 58.3 ± 12.9 years. Twelve patients (70.6%) were diagnosed with glioblastoma; two patients (11.8%) with anaplastic astrocytoma and three patients (17.6%) with anaplastic oligodendroglioma. Mean time interval between MRP and BOLD-CVR was 6.4 days (0–31) and mean tumor volume was 33.58 ± 49.37 cm³ (Table 1).

3.2. BOLD-CVR & MRP findings

Comparison between values in the affected and unaffected hemispheres showed marked BOLD-CVR impairment inside the lesion up to 21 mm outside the contrast enhancing lesion (i.e., seventh perilesional VOI, $p < 0.05$; see Table 2). Fig. 4 shows the peritumoral BOLD-CVR evolution within the 3 mm VOIs moving away from the tumor lesion. Here increasing CVR values from the tumor to the outer VOIs can be observed. Fig. 5 depicts the MRP findings in the same VOIs of the affected hemisphere.

Finally, IDH-status and WHO grade did not have a relevant influence on CVR and MRP values, whereas tumor volume affected CVR values inside the lesion and in the first three VOIs ($p < 0.05$; see also Supplemental material). Pearson coefficient showed positive spatial correlations including all VOIs between CVR and rCBV ($r = 0.27$; $p < 0.001$) and rCBF ($r = 0.42$; $p < 0.001$), as well as a negative correlation between CVR and TTP ($r = -0.47$; $p < 0.001$), values of the affected hemisphere (Table 3 & Fig. 5).

4. Discussion

In this study, we demonstrate that peritumorally impaired BOLD-CVR is associated with concomitant hemodynamic changes as reflected by altered multiparametric perfusion imaging parameters (rCBF, rCBV, MTT & TTP). These hemodynamic alterations are most pronounced closest to the contrast-enhancing tumor lesion with the severity of CVR

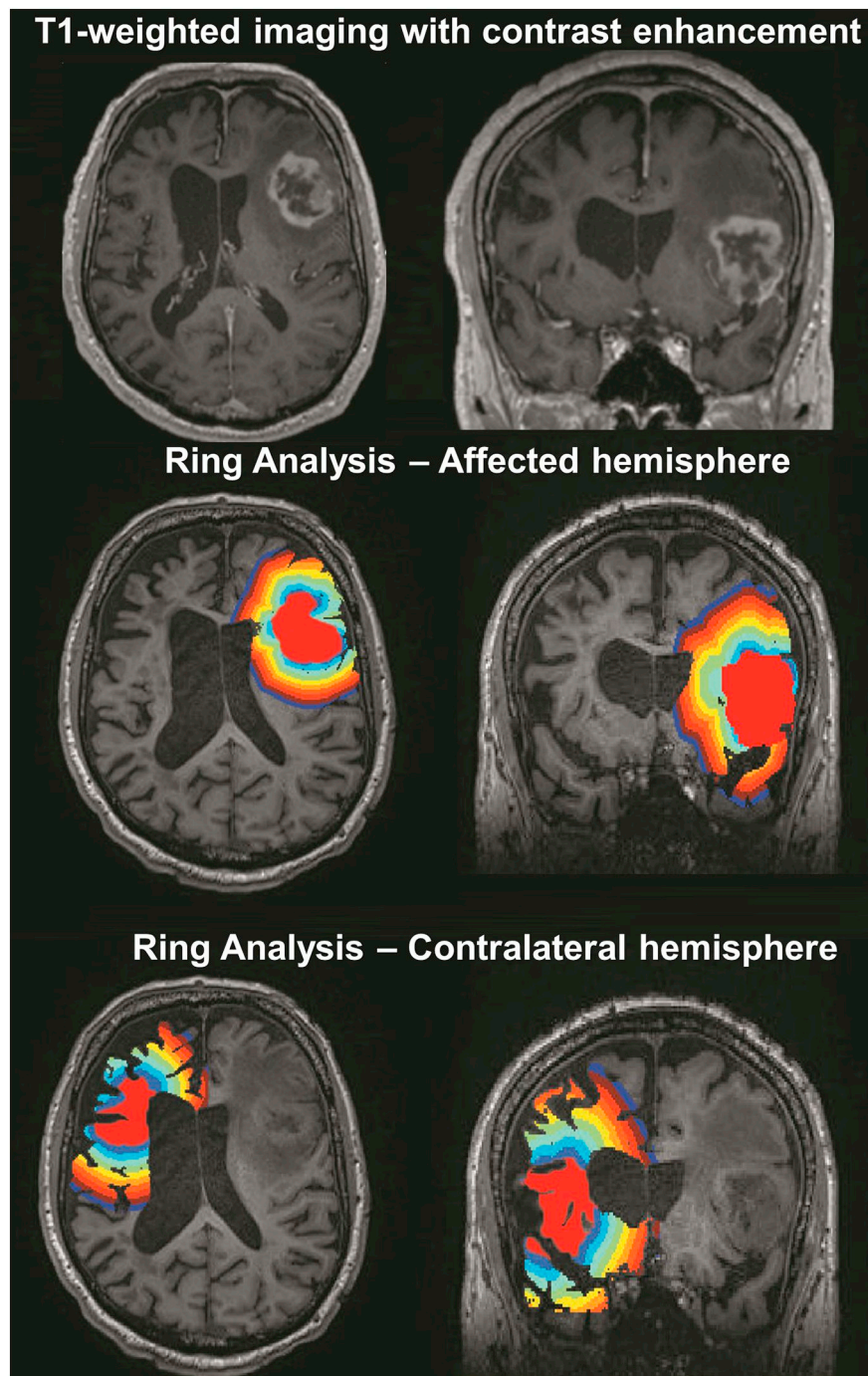


Fig. 3. Illustrative depiction of 3D concentric expanding VOI analysis. Cerebrovascular reactivity (CVR) and magnetic resonance perfusion (MRP) values were calculated in the tumor (as identified on T1-weighted contrast-enhanced MR – the red region in the picture) and in concentric volumes of interest (VOI) outside the tumor. Each VOI has a 3 mm width. Values were obtained until VOI number 10 (= 30mm from the tumor border). The same analysis was performed for the unaffected hemisphere by flipping the same VOIs to the contralateral side.

impairment correlating to tumor volume. This finding fits earlier findings that tumor volume is correlated to the whole brain and hemispheric CVR impairment. The distribution of these multiparametric hemodynamic MRI patterns may be considered for future studies characterizing the hemodynamic peritumoral environment, thereby better identifying the extent of tumor infiltration.

4.1. BOLD-CVR in diffuse gliomas

The first application of BOLD fMRI in cerebral gliomas was to

discriminate “eloquent” areas from tumor tissue, in order to perform safe and more extended surgical resection [18,19]. Here, functionally activated brain areas are depicted due to a relative decrease in deoxyhemoglobin blood concentration, following an overshoot of CBF after neuronal activity, thereby producing a change in the BOLD signal [20,21]. In brain tumors, this mechanism may be altered, i.e., neurovascular uncoupling can occur.

Contemporary BOLD studies have shown that brain tumors close to eloquent areas exhibit more often neurovascular uncoupling [22]. In this regard, since brain areas exhibiting impaired BOLD-CVR have been

Table 2

CVR comparison (paired *t*-test) between mean values calculated separately in each VOI in the affected and contralateral hemisphere. P-values were adjusted by a Benjamini-Hochberg procedure with a false discovery rate and $\alpha = 0.05$. Significant values are highlighted with bold digits. The concentric 3 mm volumes represent the ten individual expanding VOIs starting from the contrast-enhancing lesion up to 30 mm outwards.

	CVR affected hemisphere (mean $\pm$ standard deviation)	CVR contralateral hemisphere (mean $\pm$ standard deviation)	p-Value	Adjusted α
Tumor	0.05 $\pm$ 0.08	0.11 $\pm$ 0.08	< 0.001	0.005
Volume 1 (3 mm)	0.07 $\pm$ 0.09	0.12 $\pm$ 0.09	< 0.001	0.009
Volume 2 (6 mm)	0.09 $\pm$ 0.09	0.12 $\pm$ 0.09	0.004	0.014
Volume 3 (9 mm)	0.10 $\pm$ 0.09	0.12 $\pm$ 0.09	0.008	0.018
Volume 4 (12 mm)	0.11 $\pm$ 0.09	0.12 $\pm$ 0.09	0.022	0.032
Volume 5 (15 mm)	0.11 $\pm$ 0.08	0.13 $\pm$ 0.09	0.012	0.023
Volume 6 (18 mm)	0.12 $\pm$ 0.09	0.13 $\pm$ 0.10	0.029	0.036
Volume 7 (21 mm)	0.12 $\pm$ 0.09	0.13 $\pm$ 0.09	0.019	0.027
Volume 8 (24 mm)	0.13 $\pm$ 0.09	0.13 $\pm$ 0.09	0.053	0.041
Volume 9 (27 mm)	0.13 $\pm$ 0.09	0.13 $\pm$ 0.09	0.219	0.045
Volume 10 (30 mm)	0.14 $\pm$ 0.09	0.13 $\pm$ 0.09	0.746	0.050

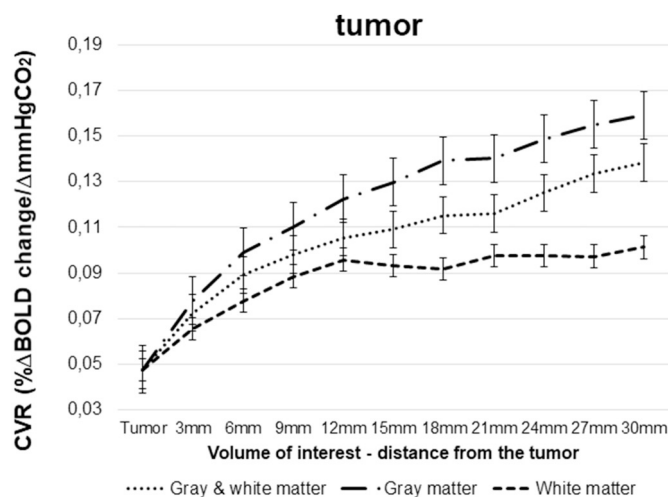


Fig. 4. Mean CVR values for every VOI in the affected hemisphere. Every volume (VOI) has a 3mm width. Standard errors of the mean are reported with brackets (CVR: cerebrovascular reactivity; BOLD: blood oxygen-level dependent).

strongly associated with neurovascular uncoupling, impaired BOLD-CVR has been suggested as a surrogate imaging marker [23–25]. Our finding may be an explanation for the increased presence of neurovascular uncoupling in tumors more adjacent to eloquent areas.

4.2. Impaired BOLD-CVR

The BOLD signal depends on blood flow, hematocrit and cerebral metabolic consumption rate of oxygen [6,20,21] and can be impaired in patients with diffuse gliomas for various reasons. The pathological tumor vasculature does not have the physiological capacity to regulate blood flow by changing its caliber (CVR), due to its abnormal structure [26–28]. This phenomenon is known as tumor neo-angiogenesis [29]. Hemodynamic components of tumor neo-angiogenesis are increased CBF and CBV. These features are more characteristics of diffuse glioma and a significant component of MRP capacity to determine pathological vascularization. When a vasoactive stimulus is applied, the abnormal tumor vessels will not respond adequately resulting in a diversion of cerebral blood flow to regions with preserved CVR. This may result in impaired or negative BOLD signal changes in the affected brain regions [26,30]. Our data seem to follow these findings.

Second, the relative hypermetabolism and hypervolemia inside the lesion can cause the peritumoral vessels to drop their perfusion pressure (i.e., dilate) in order to recruit sufficient blood flow. The consequence is impaired CVR due to exhausted reserve capacity if further vasodilation is provoked [2]. Furthermore, despite histologically intact vasculature, neoplastic cells can alter the interaction between astrocytes and vessels

biasing the neurovascular coupling mechanism and impairing the vessels' ability to regulate blood flow, as previously demonstrated [26].

Impaired CVR is present in diffuse gliomas both within the contrast-enhancing lesion and areas beyond the tumor borders. [2,4,5,23,26,29,30]. With this proposed VOI analysis method, BOLD-CVR may be able to discriminate areas of tumor invasion from healthy brain tissue.

These BOLD-CVR features could be integrated with MRP in a multimodal imaging assessment to indicate zones with altered vascular physiology. As observed, BOLD-CVR, by analyzing the whole tumor and the perilesional zone within 30 mm without the operator needing to select a VOI, was able to detect relevant differences between areas with impaired hemodynamic state and unaffected areas on the contralateral hemisphere. Potential advantages of BOLD-CVR compared to MRP are the more quantitative nature of BOLD-CVR, as well as the higher resolution of the imaging. Previous BOLD-CVR studies have shown a high intra- and intersubject agreement for BOLD-CVR studies making it an optimal imaging tool for sequential imaging assessment [31]. MRP exploits the altered tumor vasculature [32] characterized by immature and unorganized vessels with higher permeability and hyperperfusion [33] to differentiate between healthy and pathological tissue.

Although MRP has shown interesting results in the neuro-oncological field [9,11,34–37] by its own MRP has marked limitations: first, the functional parameters obtained (rCBV, rCBF, MTT, and TTP) are semi-quantitative and absolute values can vary over time in the same subject even without tumor progression or relevant alterations [35]. Second, acquisition and data processing vary among different institutions, therefore lacking a standardized protocol [35]. Third, given the considerable heterogeneity of cerebral gliomas, the advantage of using MRP is limited to areas with solid, non-necrotic regions [35]. Finally, being based on the artificial selection of VOIs, the results are highly operator-dependent, and cut-off values have to be applied for attempting to discriminate between tumor and healthy tissue [38]. Despite attempts to standardize imaging acquisition and processing among centers [32], these factors may currently lead to a false negative analysis of infiltrating tumor zones.

4.3. Limitations

Some limitations have to be mentioned for this study. The small number of patients limits the generalizability of our findings to a larger population. This could have influenced our results, particularly regarding the role of IDH-status and tumor grade concerning BOLD-CVR and MRP values. The influence of IDH-status on vascularity and CVR is a point worth investigating in the future, as IDH is known to play a relevant role in tumor metabolism [39,40] and could, therefore, significantly alter BOLD-CVR. In order to better test the potentials of BOLD-CVR, a stratification of these patients, for instance, a differentiation of WHO grade III and grade IV, would be optimal.

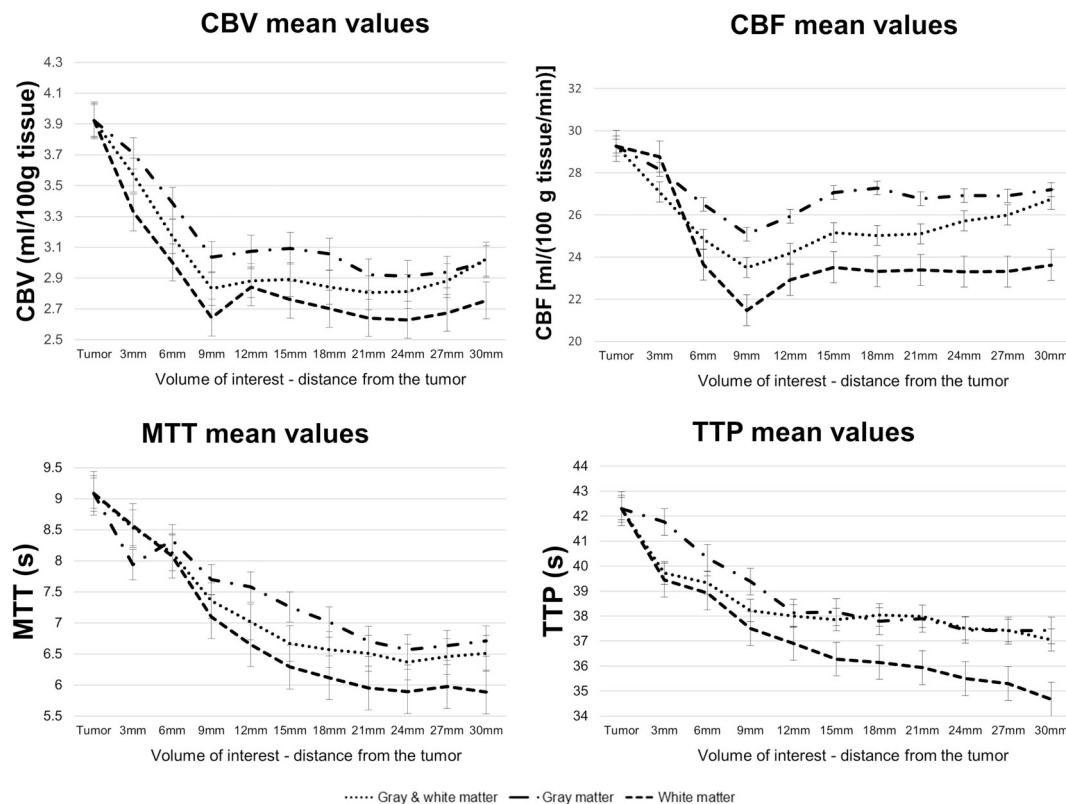


Fig. 5. MRP values in the tumor and the concentric VOIs. Values reported are means for every VOI. Standard errors of the mean are reported with brackets (CBF: cerebral blood flow; CBV: cerebral blood volume; MTT: mean transit time; TTP: time to peak; VOIs: regions of interest).

Table 3

Pearson's correlations between mean CVR and MRP values in the analyzed VOIs. P-values were considered significant after correcting for multiple comparisons (Benjamini, Krieger & Yekutieli procedure) at false discovery rate of 0.05 (unadjusted correlation critical p-value = 0.01; tumor volume adjusted correlation critical p-value = 0.001).

	Unadjusted		Tumor volume adjusted	
	r	p-Value	r	p-Value
CVR-rCBF	0.33	<0.001	0.42	<0.001
CVR-rCBV	0.18	0.01	0.27	<0.001
CVR-MTT	-0.15	0.04	-0.16	0.03
CVR-TTP	-0.40	<0.001	-0.47	<0.001

5. Conclusion

Peritumorally impaired BOLD-CVR is associated with concomitant hemodynamic alterations with severity correlating to tumor volume. The distribution of these multiparametric hemodynamic MRI patterns may be considered for future studies characterizing the hemodynamic peritumoral environment, thereby better identifying the extent of tumor infiltration.

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.mri.2020.03.012>.

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References

- [1] Fierstra J, van Niftrik B, Piccirelli M, Burkhardt JK, Pangalu A, Kocian R, et al. Altered intraoperative cerebrovascular reactivity in brain areas of high-grade glioma recurrence. *Magn Reson Imaging* 2016;34(6):803–8.
- [2] Fierstra J, van Niftrik B, Piccirelli M, Bozinov O, Pangalu A, Krayenbuhl N, et al. Diffuse gliomas exhibit whole brain impaired cerebrovascular reactivity. *Magn Reson Imaging* 2018;45:78–83.
- [3] Pillai JJ, Zaca D. Comparison of BOLD cerebrovascular reactivity mapping and DSC MR perfusion imaging for prediction of neurovascular uncoupling potential in brain tumors. *Technol Cancer Res Treat* 2012;11(4):361–74.
- [4] Iranmahboob A, Peck KK, Brennan NP, Karimi S, Fiscaro R, Hou B, et al. Vascular reactivity maps in patients with gliomas using breath-holding BOLD fMRI. *J Neuroimaging* 2016;26(2):232–9.
- [5] Holodny AI, Schulder M, Liu WC, Wolko J, Maldjian JA, Kalnin AJ. The effect of brain tumors on BOLD functional MR imaging activation in the adjacent motor cortex: implications for image-guided neurosurgery. *AJNR Am J Neuroradiol* 2000;21(8):1415–22.
- [6] Davis TL, Kwong KK, Weisskoff RM, Rosen BR. Calibrated functional MRI: mapping the dynamics of oxidative metabolism. *Proc Natl Acad Sci U S A* 1998;95(4):1834–9.
- [7] Sebok M, van Niftrik B, Piccirelli M, Bozinov O, Wegener S, Esposito G, et al. BOLD cerebrovascular reactivity as a novel marker for crossed cerebellar diaschisis.

- Neurology 2018;91(14):e1328–37.
- [8] Hendrik Bas van Niftrik C, Sebok M, Muscas G, Piccirelli M, Serra C, Krayenbuhl N, et al. Characterizing ipsilateral thalamic diaschisis in symptomatic cerebrovascular steno-occlusive patients. *J Cereb Blood Flow Metab* 2020;40(3):563–73. [271678x19830532].
 - [9] Zhang J, Liu H, Tong H, Wang S, Yang Y, Liu G, et al. Clinical applications of contrast-enhanced perfusion MRI techniques in gliomas: recent advances and current challenges. *Contrast Media Mol Imaging* 2017;2017:7064120.
 - [10] Kim M, Kim HS. Emerging techniques in brain tumor imaging: what radiologists need to know. *Korean J Radiol* 2016;17(5):598–619.
 - [11] Abrigo JM, Fountain DM, Provenzale JM, Law EK, Kwong JS, Hart MG, et al. Magnetic resonance perfusion for differentiating low-grade from high-grade gliomas at first presentation. *Cochrane Database Syst Rev* 2018;1:Cd011551.
 - [12] Patel P, Baradaran H, Delgado D, Askin G, Christos P, John Tsiouris A, et al. MR perfusion-weighted imaging in the evaluation of high-grade gliomas after treatment: a systematic review and meta-analysis. *Neuro Oncol* 2017;19(1):118–27.
 - [13] Saini J, Gupta PK, Sahoo P, Singh A, Patir R, Ahlawat S, et al. Differentiation of grade II/III and grade IV glioma by combining “T1 contrast-enhanced brain perfusion imaging” and susceptibility-weighted quantitative imaging. *Neuroradiology* 2018;60(1):43–50.
 - [14] van Niftrik CHB, Piccirelli M, Bozinov O, Pangalu A, Fisher JA, Valavanis A, et al. Iterative analysis of cerebrovascular reactivity dynamic response by temporal decomposition. *Brain Behav* 2017;7(9):e00705.
 - [15] Fierstra J, Conklin J, Krings T, Slessarev M, Han JS, Fisher JA, et al. Impaired perinidal cerebrovascular reserve in seizure patients with brain arteriovenous malformations. *Brain* 2011;134(Pt 1):100–9.
 - [16] Benjamini Y, Hochberg Y. Controlling the false discovery rate: a practical and powerful approach to multiple testing. *J R Stat Soc B Methodol* 1995;57(1):289–300.
 - [17] Krieger AM, Yekutieli D, Benjamini Y. Adaptive linear step-up procedures that control the false discovery rate. *Biometrika* 2006;93(3):491–507.
 - [18] Feldman SC, Chu D, Schulder M, Pawar R, Barry M, Cho E, et al. The blood oxygen level-dependent functional MR imaging signal can be used to identify brain tumors and distinguish them from normal tissue. *AJNR Am J Neuroradiol* 2009;30(2):389–95.
 - [19] Huang H, Lu J, Wu J, Ding Z, Chen S, Duan L, et al. Tumor tissue detection using blood-oxygen-level-dependent functional MRI based on independent component analysis. *Sci Rep* 2018;8(1):1223.
 - [20] Fierstra J, Sobczyk O, Battisti-Charbonney A, Mandell DM, Poubanc J, Crawley AP, et al. Measuring cerebrovascular reactivity: what stimulus to use? *J Physiol* 2013;591(23):5809–21.
 - [21] Fraga de Abreu VH, Peck KK, Petrovich-Brennan NM, Woo KM, Holodny AI. Brain tumors: the influence of tumor type and routine MR imaging characteristics at BOLD functional MR imaging in the primary motor gyrus. *Radiology* 2016;281(3):876–83.
 - [22] Fang S, Liang J, Qian T, Wang Y, Liu X, Fan X, et al. Anatomic location of tumor predicts the accuracy of motor function localization in diffuse lower-grade gliomas involving the hand knob area. *Am J Neuroradiol* 2017;38:1990–7.
 - [23] Pillai JJ, Zaca D. Clinical utility of cerebrovascular reactivity mapping in patients with low grade gliomas. *World J Clin Oncol* 2011;2(12):397–403.
 - [24] Pillai JJ, Mikulis DJ. Cerebrovascular reactivity mapping: an evolving standard for clinical functional imaging. *AJNR Am J Neuroradiol* 2015;36(1):7–13.
 - [25] van Niftrik CHB, Piccirelli M, Bozinov O, Maldaner N, Strittmatter C, Pangalu A, et al. Impact of baseline CO₂ on blood-oxygenation-level-dependent MRI measurements of cerebrovascular reactivity and task-evoked signal activation. *Magn Reson Imaging* 2018;49:123–30.
 - [26] Pak RW, Hadjiabadi DH, Senarathna J, Agarwal S, Thakor NV, Pillai JJ, et al. Implications of neurovascular uncoupling in functional magnetic resonance imaging (fMRI) of brain tumors. *J Cereb Blood Flow Metab* 2017;37(11):3475–87.
 - [27] Jain RK. Normalization of tumor vasculature: an emerging concept in anti-angiogenic therapy. *Science* 2005;307(5706):58–62.
 - [28] Bullitt E, Zeng D, Gerig G, Aylward S, Joshi S, Smith JK, et al. Vessel tortuosity and brain tumor malignancy: a blinded study. *Acad Radiol* 2005;12(10):1232–40.
 - [29] Agarwal S, Sair HI, Yahyavi-Firouz-Abadi N, Airan R, Pillai JJ. Neurovascular uncoupling in resting state fMRI demonstrated in patients with primary brain gliomas. *J Magn Reson Imaging* 2016;43(3):620–6.
 - [30] Sobczyk O, Battisti-Charbonney A, Fierstra J, Mandell DM, Poubanc J, Crawley AP, et al. A conceptual model for CO₂-induced redistribution of cerebral blood flow with experimental confirmation using BOLD MRI. *Neuroimage* 2014;92:56–68.
 - [31] Kassner A, Thornhill RE. Texture analysis: a review of neurologic MR imaging applications. *AJNR Am J Neuroradiol* 2010;31(5):809–16.
 - [32] Anzalone N, Castellano A, Cadioli M, Conte GM, Cuccarini V, Bizzi A, et al. Brain gliomas: multicenter standardized assessment of dynamic contrast-enhanced and dynamic susceptibility contrast MR images. *Radiology* 2018;287(3):933–43. 170362.
 - [33] Lu-Emerson C, Duda DG, Emblem KE, Taylor JW, Gerstner ER, Loeffler JS, et al. Lessons from anti-vascular endothelial growth factor and anti-vascular endothelial growth factor receptor trials in patients with glioblastoma. *J Clin Oncol* 2015;33(10):1197–213.
 - [34] Fathi Kazerooni A, Nabil M, Zeinali Zadeh M, Firouznia K, Azmoudeh-Ardalan F, Frangi AF, et al. Characterization of active and infiltrative tumorous subregions from normal tissue in brain gliomas using multiparametric MRI. *J Magn Reson Imaging* 2018;48(4):938–50.
 - [35] Delgado AF, Delgado AF. Discrimination between glioma grades II and III using dynamic susceptibility perfusion MRI: a meta-analysis. *AJNR Am J Neuroradiol* 2017;38(7):1348–55.
 - [36] Delgado-Lopez PD, Rinones-Mena E, Corrales-Garcia EM. Treatment-related changes in glioblastoma: a review on the controversies in response assessment criteria and the concepts of true progression, pseudoprogression, pseudoreponse and radionecrosis. *Clin Transl Oncol* 2018;20(8):939–53.
 - [37] Tensaouti F, Khalifa J, Lusque A, Plas B, Lotterie JA, Berry I, et al. Response Assessment in Neuro-Oncology criteria, contrast enhancement and perfusion MRI for assessing progression in glioblastoma. *Neuroradiology* 2017;59(10):1013–20.
 - [38] Dhermain FG, Hau P, Lanfermann H, Jacobs AH, van den Bent MJ. Advanced MRI and PET imaging for assessment of treatment response in patients with gliomas. *Lancet Neurol* 2010;9(9):906–20.
 - [39] Venneti S, Thompson CB. Metabolic reprogramming in brain tumors. *Annu Rev Pathol Mech Dis* 2017;12(1):515–45.
 - [40] Cohen AL, Holmen SL, Colman H. IDH1 and IDH2 mutations in gliomas. *Curr Neurol Neurosci Rep* 2013;13(5):345.